

The Spinor Residue: $g = 2$ and Stern–Gerlach without Relativity

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Abstract

The charges-as-configurations reading of RealQM removes relativity and gives the electron’s magnetic moment as a quantised current-winding: an orbital winding $e^{im\phi}$ carries $\mu_z = -m\mu_B$ with gyromagnetic factor $g = 1$. The key is that “spin” bundles two logically distinct things: the fermionic *exclusion*, which RealQM realises geometrically by non-overlapping domains with no spin variable (and which accounts for the periodic table, shells, and bonding), and the two-valued *magnetic* moment, which it does not yet carry. This leaves exactly one residue — the electron’s *spin* moment has $g = 2$, twice the orbital value per unit angular momentum — and one experiment that RealQM in its present form cannot reproduce: *Stern–Gerlach*, whose two-spot splitting is the signature of two-valued spin $\frac{1}{2}$. We make the residue precise, and argue that it is *not* a demand for relativity. By Lévy-Leblond’s theorem $g = 2$ follows from a first-order (linearised), *non-relativistic* spinor wave equation; the “2” is the SU(2) double cover of the rotation group, a geometric fact, not a consequence of Lorentz invariance. Concretely, writing the kinetic energy with $\boldsymbol{\sigma} \cdot \mathbf{p}$ rather than p^2 produces, in a field, the term $-q\boldsymbol{\sigma} \cdot \mathbf{B}$ — the $g = 2$ Zeeman coupling — automatically. The target for RealQM is therefore identified without importing relativity: equip the non-overlapping domains with the $\pm\frac{1}{2}$ spinor label the base theory already suggests, implemented as a $\boldsymbol{\sigma} \cdot \mathbf{p}$ structure, so that the moment becomes two-valued ($\mu_z = \pm\mu_B$) and Stern–Gerlach’s two spots and $g = 2$ follow together. We are explicit that this is a *target*, not a completed derivation: what is established is that the residue is geometric and reachable, not relativistic.

1 The one residue

In the current-bearing form of RealQM the magnetic moment is that of the charge current, $\boldsymbol{\mu} = \frac{1}{2} \int \mathbf{r} \times \mathbf{J}$, and an orbital phase winding $e^{im\phi}$ gives the quantised value $\mu_z = -m\mu_B$ [1, 2]. This is gyromagnetic factor $g = 1$: one unit of orbital angular momentum, $L_z = m\hbar$, yields m Bohr magnetons. The charges-as-configurations picture [3] builds on this to remove any need for relativity: sizes and moments are properties of Coulomb configurations, a “free” electron is only a weakly-confined one, and the winding it can afford gives μ_B with no rest-energy scale invoked.

Exactly one thing is left unaccounted. The electron’s *spin* moment is not $g = 1$ but $g = 2$: a spin of only $\hbar/2$ produces a *full* Bohr magneton, twice what an orbital current of the same angular

momentum would give. An orbital winding cannot supply this factor of two; it is the signature of spin, and RealQM has, so far, no spin- $\frac{1}{2}$ moment. This is the residue.

2 What spin is: exclusion versus magnetism

That the residue is so small has a reason worth stating in its own right, because it clarifies what RealQM does and does not owe to spin. The word “spin” bundles together two *logically distinct* things:

1. **Statistics-spin** — the fermionic *exclusion*: two electrons to an orbital, antisymmetry, the Pauli principle, shell structure, the periodic table.
2. **Magnetic-spin** — the two-valued *magnetic moment*: $\mu_z = \pm\mu_B$, $g = 2$, the property that splits a beam.

The spin–statistics theorem welds them (“half-integer spin $\Rightarrow$ fermion”), so in the standard formulation they appear as one property. They are not, and RealQM separates them.

Statistics-spin is geometry, not spin. Everything RealQM reproduces — atoms, the periodic table, shells, bonding, reactions, condensed phases, nuclei — follows from *non-overlapping domains* in place of antisymmetry: “two electrons per orbital” becomes simply “two spatial territories,” with no spin variable anywhere, and nothing missing. So the exclusion half of spin is *reducible to geometry*; it was doing bookkeeping that a spatial constraint does without it. This is why the whole of the programme has met no need for spin: everything built so far is the geometric half.

Magnetic-spin is irreducible. What is left over is the two-valued moment — observed directly and inescapably in Stern–Gerlach, in $g = 2$ to twelve digits, and in the everyday technology of EPR, NMR, and MRI. This is not bookkeeping; it is a physical degree of freedom, and it is precisely what RealQM does not yet carry. The residue is therefore *small and sharp*: not “spin” in its entirety, but only its SU(2) magnetic factor of two.

A conjecture on the connection — and a check that tempers it. If exclusion is geometric and the magnetic moment is a separate SU(2) structure, then the spin–statistics *connection* itself might seem an artefact of bundling the two under one name. It is tempting to conjecture so. But a test weighs against the strong form: closed-shell *diamagnetism* (§6) requires the two electrons’ spins to pair antiparallel, and geometric non-overlap alone does not force this — so the connection reappears, as needed, precisely where more than one electron is magnetically active. We therefore state the position carefully: RealQM decouples exclusion from magnetism at the *single-electron* level (the caged moment vanishes, the free moment is a winding), but at the *many-electron* level the pairing that statistics enforces is still required for correct magnetism. Whether that pairing can be supplied *geometrically* within RealQM (a shared-boundary coupling) or must be imported as a spin-statistics term is the open question; the strong conjecture that the weld is mere convention is not supported by the closed-shell test.

3 Its experimental face: Stern–Gerlach

The residue is not academic; it is the whole content of the Stern–Gerlach experiment. A beam of silver atoms — each with a single valence electron in an $L = 0$ (orbitally spherical) ground state — passes through an inhomogeneous magnetic field and splits into *two* discrete spots. Two spots is the signature of a two-valued moment, $\mu_z = \pm\mu_B$, i.e. spin $\frac{1}{2}$ with $g = 2$; an orbital moment would split into an *odd* number $2l + 1$, and for the $L = 0$ silver electron would not split at all ($m = 0$, $\mu_z = 0$).

RealQM in its present form therefore *fails* Stern–Gerlach: with only orbital windings the $L = 0$ valence electron carries no moment and the beam would pass undeflected. Reproducing the two-spot pattern requires precisely the two-valued spin- $\frac{1}{2}$ structure that is missing. Stern–Gerlach and $g = 2$ are one problem: the two spots *are* $g = 2$ made visible.

4 What a charge density needs: two with no middle

It is worth pinning down exactly what a magnetic field asks of a charge density, because it isolates the residue to a single feature. A field couples to charge only through *motion* — through the current $\mathbf{J}$ — and three levels follow:

- **To react at all**, a density needs a current. A static real density ($\mathbf{J} = 0$) is magnetically inert to first order; a field exerts force and shifts energy only through $\mathbf{J}$ (the coupling $-\frac{q}{m}\mathbf{J} \cdot \mathbf{A}$, or the Lorentz $\mathbf{J} \times \mathbf{B}$). The one exception is diamagnetism — a weak *induced* current — which is uniform and does not split a beam.
- **To carry a moment**, the current must *circulate*: a phase winding $e^{im\phi}$ gives $\mu_z = -m\mu_B$. This a scalar RealQM density does whenever it has orbital circulation.
- **To split into two**, as in Stern–Gerlach, is the one thing a scalar circulation cannot do. A scalar winding admits $m = 0, \pm 1, \pm 2, \dots$ — an *odd* ladder that *includes zero*. For $|m| \leq 1$ it gives three orientations ($-\mu_B, 0, +\mu_B$), three spots, with an undeflected *middle* beam. Stern–Gerlach gives two spots, $\pm\mu_B$, *with no middle*. That missing zero is the whole signature: it is the half-integer structure $m_s = \pm\frac{1}{2}$ (no $m_s = 0$), which a scalar density cannot produce.

So the residue reduces to a single phrase: **two with no middle**. Reacting to a field (a current) and carrying a moment (circulation) are ordinary moving-charge electromagnetism, already present in scalar RealQM; the irreducible extra is only the two-valued, zero-free structure of a half-integer spinor — the SU(2) feature, and nothing more. What Stern–Gerlach forbids is making that two-valuedness *vanish*; what it leaves open is whether the structure must be *posited* (as in the Dirac account) or can *emerge* from a two-valued, node-forced feature of a circulating configuration — the distinction between eliminating spin and *deriving* it (§6).

5 The residue is geometric, not relativistic

It is tempting to conclude that spin, $g = 2$, and Stern–Gerlach force relativity into the theory, since in the textbook account $g = 2$ falls out of the Dirac equation. This conclusion is false, and the

point is decisive for the present programme.

Lévy-Leblond. In 1967 Lévy-Leblond showed that $g = 2$ follows from a first-order (linearised) *non-relativistic*, Galilean spinor wave equation [4]. Relativity is sufficient to produce $g = 2$ but not necessary; the factor of two survives the non-relativistic limit because it does not come from Lorentz invariance in the first place.

The mechanism. Write the kinetic energy not as $p^2/2m$ but as $(\boldsymbol{\sigma} \cdot \mathbf{p})^2/2m$ — legal for free motion since $(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = p^2$. In a magnetic field, minimal coupling $\mathbf{p} \rightarrow \mathbf{p} - q\mathbf{A}$ gives

$$\frac{(\boldsymbol{\sigma} \cdot (\mathbf{p} - q\mathbf{A}))^2}{2m} = \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} - \frac{q}{2m} \boldsymbol{\sigma} \cdot \mathbf{B}, \quad (1)$$

and the last term is exactly the $g = 2$ spin–Zeeman coupling, appearing *automatically* the moment the dynamics is linearised in $\boldsymbol{\sigma} \cdot \mathbf{p}$. No relativity is used in (1); it is the ordinary Pauli Hamiltonian.

The geometry. The “2” is the **SU(2) double cover of the rotation group**: a spinor returns to itself only after 720° , rotating at half angle. That half is reciprocally the double in the moment. It is the same species of fact as the winding-number topology that fixes $\mu_z = -m\mu_B$ in the orbital case — a statement about how spin- $\frac{1}{2}$ objects transform, geometric and scale-free, owing nothing to rest energy.

6 The target for RealQM

The residue thus lands on the reachable side of the ledger. What RealQM needs is not relativity but the right *spinor structure* on its domains, and the base theory already points at it: the ψ_i on Ω_i acquiring an internal $\pm\frac{1}{2}$ label. The programme is to implement that label not as a bolt-on but as a $\boldsymbol{\sigma} \cdot \mathbf{p}$ linearisation of the one-electron dynamics on each domain, so that:

- the spin–Zeeman term $-\frac{q}{2m} \boldsymbol{\sigma} \cdot \mathbf{B}$ of (1) appears, giving $g = 2$;
- the moment becomes *two-valued*, $\mu_z = \pm\mu_B$, so that an $L = 0$ electron in an inhomogeneous field splits into *two* beams — Stern–Gerlach — rather than none;
- the non-overlap/free-boundary structure, and the current $\mathbf{J}$ of the complex form, are retained unchanged, since (1) is a rewriting of the kinetic term, not a new interaction.

A first implementation (single electron). The single-electron step has been carried out numerically. On a two-dimensional grid a two-component spinor $\psi = (\psi_\uparrow, \psi_\downarrow)$ is evolved with the $\boldsymbol{\sigma} \cdot \mathbf{D}$ structure, $\mathbf{D} = \mathbf{p} - q\mathbf{A}$, in a uniform field. Two facts come out. First, the operator identity $(\boldsymbol{\sigma} \cdot \mathbf{D})^2 = \mathbf{D}^2 \mathbb{I} - q \boldsymbol{\sigma} \cdot \mathbf{B}$ holds to grid accuracy (residual $\sim 10^{-3}$, i.e. $O(dx^2)$), so the $g = 2$ Zeeman term $-q \boldsymbol{\sigma} \cdot \mathbf{B}$ *emerges* from squaring $\boldsymbol{\sigma} \cdot \mathbf{D}$ — it is not inserted by hand. Second, an $L = 0$ (orbitally inert) state — the very state a scalar density leaves undeflected — splits under the spinor into *two* levels $\pm\mu_B B$, splitting $2\mu_B B$, $g_s = 2$, with *no middle*, against the three-level ($m = -1, 0, +1$) orbital pattern. This is Stern–Gerlach: the $\boldsymbol{\sigma} \cdot \mathbf{p}$ structure delivers exactly the

residue, non-relativistically. The distinguishing content is that spin reaches $\pm\mu_B$ at $\pm\frac{1}{2}$ unit of angular momentum ($g = 2$, no middle) where orbital needs ± 1 unit ($g = 1$, with middle).

We state plainly what remains a *target*. Two things are not yet settled. (i) *Many-electron consistency* — and here the naive extension fails. The spinor is a per-electron magnetic label, and it must coexist with the geometric realisation of Pauli exclusion by non-overlap *without double-counting the spin* — concretely, a filled shell of two electrons must pair to zero net moment (diamagnetism). We tested the *spin sector* of the naive version — two electrons, each carrying an independent spinor with no coupling between them, in an external field, the spatial structure left aside — and it does *not* pass: the ground state is the aligned pair $\downarrow\downarrow$ with net moment $2\mu_B$, *paramagnetic*, where the observed closed shell is *diamagnetic*. (This is the spin bookkeeping alone; no concrete two-domain spatial system, such as the two Bernoulli half-domains of helium, was solved — which is exactly where a shared-boundary coupling might change the answer.) Nothing in geometric non-overlap forces the two spins antiparallel; obtaining diamagnetism requires an antiparallel-pairing (exchange) term $J\mathbf{S}_1\mathbf{S}_2$, which is a spin-statistics ingredient not supplied by non-overlap. Diamagnetism needs only the antiparallel *product* state $\uparrow\downarrow$ (net moment zero, not the entangled singlet), so the open question is sharp: can a *geometric* coupling — continuity of the spinor field across a shared domain boundary, forcing adjacent domains antiparallel like a domain wall — supply the pairing from geometry rather than by re-importing spin-statistics? That is unshown, and until it is, closed-shell magnetism is a genuine gap. (ii) *Why the spinor*. Using $(\boldsymbol{\sigma} \cdot \mathbf{D})^2$ rather than $\mathbf{D}^2$ is here a choice; Lévy-Leblond *derives* it from Galilean covariance of a first-order equation, but whether RealQM's domain structure *forces* it or merely permits it is open. What is established is stronger than the map, however: the residue is not only geometric and non-relativistic but *concretely delivered* for one electron by a $\boldsymbol{\sigma} \cdot \mathbf{p}$ structure on the domains, with $g = 2$ emerging rather than posited. It belongs inside RealQM, and a working single-electron piece is now in hand.

7 Conclusion

The charges-as-configurations picture closes the free-charge and relativity problems and leaves a single, sharp residue: the electron's spin factor $g = 2$, whose experimental embodiment is the two-spot Stern–Gerlach splitting. That residue is not a relativistic quantity smuggled in through the Dirac equation; by Lévy-Leblond it is a non-relativistic, geometric consequence of the SU(2) spinor structure, delivered by a $\boldsymbol{\sigma} \cdot \mathbf{p}$ linearisation. RealQM therefore has a concrete, relativity-free target for spin: give its domains the $\pm\frac{1}{2}$ label as a $\boldsymbol{\sigma} \cdot \mathbf{p}$ structure, and $g = 2$ and Stern–Gerlach's two beams should follow together. Until that is carried through the multiphase problem the point is a target, not a result — but it is the right size of target, and it is inside the theory, not beyond it.

References

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